

FOR LEARNERS - READ BEFORE DAY ONE



Student Setup Guide

Enterprise Data Engineering with SQL, Redis & Kafka

120 Hours | 12 Modules

Before day one

Work through this sheet **before the first session**. It takes about 45 minutes, and almost all of it is waiting for installers. If you arrive with this done, you start Module 1 writing SQL instead of watching a progress bar.

Everything here is free. Nothing needs a licence key.

Do not install everything today. Only Part 1 and Part 2 are needed for day one. Parts 4, 5 and 6 are needed later, and each one says which module it is for.

What you are installing, and when

Part	Software	Needed from	Install by
1	PostgreSQL 18 and a query client	Module 1	Before day one
2	The ICICI Bank database	Module 1	Before day one
3	Python 3.11	Module 7	Before day one, it is quick
4	Docker Desktop	Module 7	Before Module 7
5	SQL Server 2022 Developer and SSMS, loaded	Module 10	Before Module 10
6	Oracle XE and MySQL, both loaded	Module 11	Before Module 11

Parts 2, 5 and 6 each load the **same** ICICI Bank data into a different database. Nine tables, 841 rows, identical numbers everywhere. Only the schema file changes per database.

Part 1 - PostgreSQL

PostgreSQL is the database you will use for the first nine modules. It is free, it installs in about ten minutes, and its error messages are the clearest of any database you will meet.

Install it

1. Download the Windows installer from the EnterpriseDB download page for **PostgreSQL 18**.
2. Run it. Accept every default **except** the password screen.
3. On the password screen, set a password for the `postgres` user and **write it down**. There is no recovery. Use something you can type quickly, because you will type it many times in the next 120 hours.
4. Leave the port as **5432**.
5. When the installer offers to launch Stack Builder at the end, tick nothing and close it.

Confirm it works

Open a new PowerShell window and run:

```
psql --version
```

You should see a line beginning `psql (PostgreSQL) 18`.

Always type `-U postgres`. Run on its own, `psql` uses your **Windows account name** as the database username, and no such database user exists. You get a password error that has nothing to do with your password. Every command in this sheet includes `-U postgres` for exactly this reason, so copy them as they are written.

If you see "psql is not recognized", the installer did not add PostgreSQL to your PATH. Add `C:\Program Files\PostgreSQL\18\bin` to your PATH under System Properties, Environment Variables, Path, New. Then open a **new** PowerShell window. The window you already had open will never see the change. The same applies to `createdb`, `pg_dump` and every other PostgreSQL command, because they all live in that one folder.

Install a query client

A visual client makes execution plans far easier to read, and you will spend a whole module on execution plans.

pgAdmin comes with PostgreSQL and is already installed. **DBeaver Community** is the better choice, because one client connects to PostgreSQL, Oracle, MySQL and SQL Server, and you need all four before the programme ends. Install either one. Install DBeaver if you are unsure.

Part 2 - The ICICI Bank database

Everything you do for 120 hours runs against one bank: 8 branches, 30 customers, 36 accounts and 495 transactions. Your trainer will give you a folder called `00_Programme_Overview`. Inside it is a `sql` folder. You need that folder on your machine.

There is no separate login step

This trips up almost everyone, so read it before you type anything.

You do **not** log in first and then run commands. **Every command logs in by itself**, does its work, and logs out again. The `-U postgres` part is the login, and the password prompt is that login happening. When the command finishes you are back at the PowerShell prompt, logged out.

`psql` behaves in two completely different ways depending on whether you give it `-f`.

What you type	What happens
<code>psql -U postgres -d icicibank</code>	Connects and stays open . You get an <code>icicibank=#</code> prompt and type SQL until <code>\q</code> .
<code>psql -U postgres -d icicibank -f somefile.sql</code>	Connects, runs the file, disconnects . No prompt appears. You are returned to PowerShell.

The second form is the one used below. It gives you no prompt, so it can look as though nothing happened. That is correct behaviour, not a failure.

Create and load

Open PowerShell **in the folder that contains `sql`**. Run these three commands one at a time, all three at the PowerShell prompt. Do not start `psql` first.

```
createdb -U postgres icicibank
```

```
psql -U postgres -d icicibank -f sql/01_schema_postgres.sql
```

```
psql -U postgres -d icicibank -f sql/02_seed_postgres.sql
```

The order matters. `createdb` has to come first, because the other two connect to `icicibank` with `-d icicibank`, and a database cannot be connected to before it exists.

If `createdb` is not recognised

`createdb` is a small program that ships with PostgreSQL, and it lives in the same folder as `psql`. If PowerShell cannot find it, that folder is not on your PATH. Use the full path instead:

```
& "C:\Program Files\PostgreSQL\18\bin\createdb.exe" -U postgres icicibank
```

The SQL alternative

`createdb` is only a convenience wrapper. The database can equally be created with SQL, which is what the official PostgreSQL documentation shows. Start `psql`, create it, then leave:

```
psql -U postgres
```

```
CREATE DATABASE icicibank;
```

Then `\q` to return to PowerShell, and carry on with the two `-f` commands above.

Form	Where you run it	Same result
<code>createdb -U postgres icicibank</code>	PowerShell	Yes
<code>CREATE DATABASE icicibank;</code>	Inside psql	Yes

Neither one works in the other's prompt. `createdb` typed inside `psql` is not SQL, and `CREATE DATABASE` typed in PowerShell is not a program. This is the same two-prompts point made below, and it is worth getting straight now.

Each command asks for the password you wrote down in Part 1. Being asked three times is normal, because it is three separate logins. The first prints nothing at all. The second prints a long list of `CREATE TABLE` lines. The third prints `INSERT 0 1` many times.

Tired of typing the password? Run `$env:PGPASSWORD = "yourpassword"` once, and every command in that PowerShell window will skip the prompt. It applies only to that window and is forgotten when you close it. Convenient for a lab machine, and never something to do on a real server.

Confirm the load

Start a session and run the full check:

```
psql -U postgres -d icicibank
```

```
SELECT 'branch' AS table_name, COUNT(*) AS row_count FROM branch
UNION ALL SELECT 'customer',      COUNT(*) FROM customer
UNION ALL SELECT 'account',      COUNT(*) FROM account
UNION ALL SELECT 'account_txn',  COUNT(*) FROM account_txn
UNION ALL SELECT 'employee',     COUNT(*) FROM employee
UNION ALL SELECT 'loan',         COUNT(*) FROM loan
UNION ALL SELECT 'loan_payment', COUNT(*) FROM loan_payment
UNION ALL SELECT 'card',         COUNT(*) FROM card
UNION ALL SELECT 'beneficiary',  COUNT(*) FROM beneficiary
ORDER BY table_name;
```

Your numbers must match these **exactly**. The seed data is generated deterministically, so every person in the room has identical data. When your trainer says "you should see 4 rows", you should see 4 rows.

Table	Rows
<code>account</code>	36
<code>account_txn</code>	495

Table	Rows
beneficiary	60
branch	8
card	32
customer	30
employee	24
loan	12
loan_payment	144

Type `\q` to leave `psql`.

If your numbers are higher than the table, you ran the seed file twice. That is the single most common setup mistake on this programme. Fix it with Part 7.

Two prompts, and why it matters

You will work at two different prompts for the next 120 hours, and confusing them wastes more time than any other single thing on this sheet.

Prompt	Where you are	What it accepts
<code>C:\Users\you></code> or <code>PS C:\></code>	PowerShell	<code>psql</code> , <code>createdb</code> , <code>python</code> , <code>docker</code>
<code>icicibank=#</code>	Inside <code>psql</code>	SQL, and <code>psql</code> backslash commands
<code>icicibank-#</code>	Inside <code>psql</code> , mid-statement	It is still waiting for your <code>;</code>

The dash prompt is the one that confuses people. `icicibank-#` means `psql` has read part of a statement and is waiting for the semicolon that ends it. Press **Ctrl+C** to throw the half-finished statement away and get back to `=#`.

A command starting with `psql`, `createdb` or `pip` belongs at the PowerShell prompt. Typing it inside `psql` makes `psql` try to read it as SQL, which is how most people end up stuck at `-#`.

The `psql` commands worth memorising today

These are `psql`'s own commands, not SQL. They take no semicolon.

Command	What it does
<code>\l</code>	List every database on the server
<code>\c icicibank</code>	Connect to a different database
<code>\dt</code>	List the tables in the current database
<code>\d account</code>	Describe one table: columns, types, indexes, constraints
<code>\i sql/01_schema_postgres.sql</code>	Run a SQL file from inside <code>psql</code>
<code>\x</code>	Switch between row and column display, useful for wide results
<code>\timing</code>	Show how long each query took, used heavily from Module 5

Command	What it does
<code>\q</code>	Quit back to PowerShell

`SHOW DATABASES` **does not exist in PostgreSQL**. That is MySQL. The PostgreSQL equivalent is `\l`, and the equivalent of `SHOW TABLES` is `\dt`. If you have learned MySQL before this course, these two are the first habits to unlearn.

Fix every PATH at once

None of the four database installers adds its tools to the Windows PATH. Each one fails the same way, weeks apart, and each time it looks like a new problem:

```
'psql' is not recognized as an internal or external command  
'mysql' is not recognized as an internal or external command  
'sqlplus' is not recognized as an internal or external command  
'sqlcmd' is not recognized as an internal or external command
```

The program is installed and the server is usually running. Windows simply does not know where the program is. Add all four folders in one go and you will never see this again.

Database	Folder to add
PostgreSQL	C:\Program Files\PostgreSQL\18\bin
MySQL	C:\Program Files\MySQL\MySQL Server 8.0\bin
Oracle XE	C:\app\<your Windows username>\product\21c\dbhomeXE\bin
SQL Server	C:\Program Files\Microsoft SQL Server\Client SDK\ODBC\170\Tools\Binn

The Oracle path contains your own Windows username. It is not a typo and it is not the same on any two machines. If C:\app does not exist, search your C: drive for `sqlplus.exe` and use the folder it sits in.

Add them through System Properties, Environment Variables, Path, New, one line each. Or run this once in PowerShell, editing the Oracle line to match your machine:

```
[Environment]::SetEnvironmentVariable("Path", [Environment]::GetEnvironmentVariable("Path","User") + ";C:\Program Files\PostgreSQL\18\bin;C:\Program Files\MySQL\MySQL Server 8.0\bin;C:\app\<your Windows username>\product\21c\dbhomeXE\bin;C:\Program Files\Microsoft SQL Server\Client SDK\ODBC\170\Tools\Binn")
```

Then open a new PowerShell window and check all of them at once:

```
Get-Command psql, mysql, sqlplus
```

Four paths means you are done. An error naming one of them means that folder is wrong, so go and find the executable.

Until you fix the PATH, call the program by its full path with the `&` operator, which is how PowerShell runs a program whose path contains spaces:

```
& "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root -p
```

Part 3 - Python

Needed properly from Module 7, but install it now because it takes two minutes.

1. Install **Python 3.11 or later** from python.org.
2. On the first installer screen, tick **Add python.exe to PATH**. This box is easy to miss, and missing it causes most of the problems that follow.

```
python --version
```

```
pip install "psycpg[binary]" redis confluent-kafka
```

Part 4 - Docker Desktop, before Module 7

Redis and Kafka both run in containers. Installing them natively on Windows is painful, and Docker reduces it to a single command.

1. Install **Docker Desktop for Windows**.
2. It will ask to install the WSL 2 backend. Allow it. This needs a restart.
3. After restarting, open Docker Desktop and wait until the whale icon in the system tray stops animating.

```
docker run --rm hello-world
```

If you see "WSL 2 installation is incomplete", run `wsl --install` in PowerShell as Administrator, restart, and open Docker Desktop again. If it still fails, hardware virtualisation is switched off in your BIOS, and turning it on needs a reboot into BIOS settings. Do this well before Module 7, not on the morning of it.

Part 5 - SQL Server, before Module 10

Module 10 is twenty hours on Microsoft SQL Server, and roughly half of it depends on SQL Server Agent.

Install it

1. Install **SQL Server 2022 Developer edition**. It is free and full-featured.
2. Choose the **Basic** installation type.
3. Install **SQL Server Management Studio**, known as SSMS, separately afterwards. It is a different download.

Express edition will not work for this module. Express has no SQL Server Agent, so every scheduling, alerting and job exercise becomes impossible. Developer edition is the free one that has everything.

Load the bank

SQL Server has its **own** seed file, `02_seed_sqlserver.sql`, because its date and identity syntax differs from the others. Do not use the PostgreSQL seed here.

A default local install uses Windows authentication, which is what `-E` means below. No password is needed.

```
sqlcmd -S localhost -E -Q "CREATE DATABASE ICICIBank"
```

```
sqlcmd -S localhost -E -d ICICIBank -i sql\01_schema_sqlserver.sql
```

```
sqlcmd -S localhost -E -d ICICIBank -i sql\02_seed_sqlserver.sql
```

You can do the same thing in SSMS instead: open each `.sql` file, choose `ICICIBank` in the database dropdown, and press F5.

Confirm it with the same nine counts as Part 2:

```
SELECT 'account' AS table_name, COUNT(*) AS row_count FROM account  
UNION ALL SELECT 'account_txn', COUNT(*) FROM account_txn;
```

Module 10 builds its own database, `ICICIBankDW`. The `ICICIBank` database you just created is for the Module 1 dialect comparison. Leave both in place, and do not be surprised to see two.

Part 6 - Oracle and MySQL, before Module 11

Module 1 compares the three dialects side by side, and Module 11 migrates a database **from** Oracle **to** PostgreSQL. You can follow Module 1 by watching someone else's screen, but Module 11 needs Oracle loaded on your own machine, because your Oracle database is the thing being migrated.

1. Install **Oracle Database 21c Express Edition**, known as XE. Write down the SYS password.
2. Install **MySQL 8 Community Server**. Write down the root password.
3. Install **SSMA for Oracle**, a free Microsoft download, before Module 11.

One seed file for both

There is no `02_seed_oracle.sql` and no `02_seed_mysql.sql`, and that is deliberate, not an omission. Look inside `02_seed_postgres.sql` and you will see why:

```
INSERT INTO branch VALUES (1, 'Fort Mumbai Main', 'ICIC0000101',  
    'Mumbai', 'Maharashtra', DATE '2005-01-05');
```

Every row is a single-row `INSERT` using the ANSI `DATE '...'` and `TIMESTAMP '...'` literals, which Oracle, MySQL and PostgreSQL all accept unchanged. Only the **schema** file differs per database, because column types and constraint syntax are not portable. The file keeps its `_postgres` name because that is where it started.

Load Oracle

Oracle XE creates a pluggable database called `XEPDB1`. Connect to it as `system` with the password you set during installation.

```
sqlplus system/yourpassword@localhost:1521/XEPDB1
```

If sqlplus is not recognised, its folder is not on your PATH. Oracle installs to `C:\app\<your Windows username>\product\21c\dbhomeXE\bin`, so the path differs on every machine. See "Fix every PATH at once" earlier in this sheet.

Create the schema owner the course uses, then switch to it:

```
CREATE USER icici IDENTIFIED BY icici;  
GRANT CONNECT, RESOURCE, UNLIMITED TABLESPACE TO icici;  
CONNECT icici/icici@localhost:1521/XEPDB1
```

Run the two files. In `SQL*Plus`, `@` means "run this file", and the path is relative to where you started `sqlplus`:

```
@sql/01_schema_oracle.sql  
@sql/02_seed_postgres.sql  
COMMIT;
```

Type `COMMIT`; and mean it. Unlike PostgreSQL and MySQL, Oracle does not commit as it goes. Without that one line your 841 rows exist only inside your session and vanish the moment it ends. This is the same autocommit difference taught in Module 6, meeting you early.

Check the load, then leave with `EXIT`:

```
SELECT COUNT(*) FROM account_txn;
```

Load MySQL

```
mysql -u root -p -e "CREATE DATABASE icicibank"
```

```
mysql -u root -p icicibank -e "source sql/01_schema_mysql.sql"
```

```
mysql -u root -p icicibank -e "source sql/02_seed_postgres.sql"
```

Do not use `< to feed the file in`. Every MySQL tutorial online writes `mysql -u root -p icicibank < file.sql`, and PowerShell does not support input redirection at all. It fails with a confusing parser error. Use the `-e "source ..."` form above, or `Get-Content file.sql | mysql -u root -p icicibank`.

If `mysql` is not recognised, the MySQL bin folder is not on your PATH. It is `C:\Program Files\MySQL\MySQL Server 8.0\bin`. Fix it with the "Fix every PATH at once" section earlier in this sheet, or work around it for now: `& "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root -p -e "CREATE DATABASE icicibank"`

Check it with `SELECT COUNT(*) FROM account_txn;`, then `exit`.

Expected counts, all three databases

The same nine numbers from Part 2 apply to Oracle, MySQL and SQL Server. The dataset is identical everywhere; only the dialect changes. If Oracle shows zero rows, you forgot the `COMMIT`;

Part 7 - Starting clean

Modules 2, 4, 9 and 11 deliberately change the schema. You are **meant** to experiment and break things. This is how you put the bank back exactly as it was, and it takes under a minute.

```
psql -U postgres -d icicibank -f sql/01_schema_postgres.sql
```

```
psql -U postgres -d icicibank -f sql/02_seed_postgres.sql
```

Run the row count check from Part 2 afterwards. If the numbers match the table, you are clean.

Learn this now and use it freely. Learners who are afraid of breaking their database learn far less than learners who break it twice a day and reset.

The errors you will actually hit

These eleven failures account for nearly every setup problem on this programme. Find yours, apply the fix, carry on.

psql, createdb, mysql, sqlplus or sqlcmd is not recognized

The program is installed and the server is probably running. Windows just does not know where the program is. This is the single most common error on this programme, and it hits all four databases.

Fix every PATH in one go using the section of that name earlier in this sheet, then open a **new** PowerShell window. A window that was already open keeps the old PATH forever.

Every PostgreSQL command line tool lives in that one folder, so when `psql` is missing, `createdb`, `pg_dump`, `pg_restore` and `pg_basebackup` are all missing too. To get moving before you fix the PATH, call the tool by its full path:

```
& "C:\Program Files\PostgreSQL\18\bin\createdb.exe" -U postgres icicibank
```

A quick way to check what PowerShell can actually find:

```
Get-Command psql, createdb
```

FATAL: password authentication failed for user "your Windows name"

Look carefully at the user name in the message. If it is your **Windows** account name rather than `postgres`, your password is fine and nothing is broken. You ran `psql` with no arguments, so it used your Windows account name as the database username, and there is no database user by that name.

PostgreSQL reports a user that does not exist and a wrong password with the identical message, deliberately, so that an attacker cannot discover valid user names. That is why this one is so confusing the first time.

Name the user every time:

```
psql -U postgres -d icicibank
```

FATAL: password authentication failed for user "postgres"

The password is wrong. Check Caps Lock first, because it is Caps Lock more often than it is anything else. If you have genuinely lost the password, the fastest fix is to uninstall PostgreSQL, delete the `C:\Program Files\PostgreSQL` folder, and reinstall. Ten minutes, and it always works.

could not connect to server: Connection refused

The database is installed but not running. Press Windows and R together, type `services.msc`, find `postgresql-x64-18`, right-click it and choose Start. Set its startup type to Automatic while you are there.

createdb: database "icicibank" already exists

Not an error. You already created it. Skip to the schema command.

duplicate key value violates unique constraint "account_pkey"

You ran the seed file twice. Go to Part 7 and start clean.

No such file or directory: sql/01_schema_postgres.sql

PowerShell is not in the folder that holds `sql`. Run `Get-Location` to see where you are and `Get-ChildItem` to see what is there. Either move to the right folder, or give the full path in quotes.

The prompt changed to `-#` and nothing I type works

`psql` is waiting for the semicolon that ends a statement. Press **Ctrl+C** to discard it and return to `=#`.

This usually happens because a PowerShell command such as `psql -U postgres -d icicibank -f sql/01_schema_postgres.sql` was typed **inside** `psql`. Either type `\q` first and run it in PowerShell, or stay in `psql` and use `\i sql/01_schema_postgres.sql` instead.

ERROR: syntax error at or near "\"

You pasted a `psql` command such as `\i` or `\dt` into pgAdmin or DBeaver. Backslash commands work only inside the `psql` terminal. In a graphical client, use the client's own menus instead.

Port 5432 is already in use

You have two PostgreSQL installations, usually an older one left over from a previous course. Uninstall the old one, or install the new one on port 5433 and add `-p 5433` to every command.

The Kafka container starts and immediately exits

Docker does not have enough memory. Open Docker Desktop, go to Settings and then Resources, and raise the memory limit to at least 4 GB.

You are ready when

	Check
[]	<code>psql --version</code> prints version 18
[]	A query against <code>account_txn</code> returns a count of 495
[]	All nine row counts match the table in Part 2 exactly
[]	A query client is installed and connects to <code>icicibank</code>
[]	<code>python --version</code> prints 3.11 or later
[]	You have written your <code>postgres</code> password somewhere you will still have it in week three

If anything on this list fails, message your trainer before the session rather than during it. Setup problems are much cheaper to fix the evening before than in front of a room of thirty people.